Modeling with partial differential equations
Part IV: advection and wave equations

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Two propagation equations

In this part, we study two evolution equations

The advection equation, for a prescribed velocity field $b : \mathbb{R}^d \to \mathbb{R}^d$

$$\partial_t U + \operatorname{div}(bU) = S \qquad \text{or} \qquad \partial_t q + b \cdot \nabla q = s$$

The wave equation, with propagation speed $c > 0$

$$\partial_{t,t} u - c^2 \Delta u = f$$

We consider classical solutions: the equations and the initial and boundary conditions hold pointwise, with all the derivatives appearing in the equations continuous

We will see that, contrary to the heat equation, both equations propagate information at finite speed and do not regularize the initial condition

Outline

- Modeling advection and waves
- Characteristics and the advection equation
- Explicit formulas and qualitative properties of waves
- Numerical approximations using finite differences

Conservative form of the advection equation

Let $U(t, x)$ be an amount per unit volume. Its total amount in a fixed region $A \subset \mathbb{R}^d$ is

$$\int_A U(t, x)\, dx$$

If J is the flux and S is the production rate per unit volume, the local balance law is

$$\partial_t U + \operatorname{div} J = S$$

For a quantity carried by the prescribed velocity field b, the advective flux is $J = bU$. Hence

$$\partial_t U + \operatorname{div} (bU) = S$$

This is the conservative form of the advection equation

Non-conservative form of the advection equation

Let $q(t,x)$ be a scalar quantity per unit mass and let ρ be the mass density of the fluid. Then $U = \rho q$ is the corresponding quantity per unit volume

Mass conservation and conservation of U give

$$\partial_t \rho + \operatorname{div}(\rho b) = 0, \qquad \partial_t(\rho q) + \operatorname{div}(\rho q b) = \rho s$$

Using the product rule,

$$\partial_t(\rho q) + \operatorname{div}(\rho q b) = q\big(\partial_t \rho + \operatorname{div}(\rho b)\big) + \rho\big(\partial_t q + b \cdot \nabla q\big)$$

and we get the non-conservative form of the advection equation

$$\partial_t q + b \cdot \nabla q = s$$

When $s = 0$, the quantity is constant along each fluid trajectory

The operator $\frac{D}{Dt} = \partial_t + b \cdot \nabla$ is the material derivative

Conservative versus non-conservative form

The physical meaning of the unknown determines the equation

- amount per unit volume U: $\partial_t U + \mathrm{div}\,(bU) = S$
- amount per unit mass q: $\partial_t q + b \cdot \nabla q = s$

For an incompressible flow, $\mathrm{div}\, b = 0$, and

$$\mathrm{div}\,(bw) = b \cdot \nabla w$$

The two differential operators then coincide

For a compressible flow,

$$\mathrm{div}\,(bw) = b \cdot \nabla w + w \mathrm{div}\, b$$

The extra term accounts for the change of volume

- $\mathrm{div}\, b > 0$: expansion and dilution of a volumetric quantity
- $\mathrm{div}\, b < 0$: contraction and concentration of a volumetric quantity

Models involving advection

Pollutant concentration per unit volume

$$\partial_t U + \operatorname{div}(bU) = S$$

With molecular diffusion, Fick's law gives

$$\partial_t U + \operatorname{div}(bU) - D\Delta U = S$$

Passive scalar per unit mass, for example salinity

$$\partial_t q + b \cdot \nabla q = 0$$

Temperature in an incompressible fluid with thermal diffusion

$$\rho c_p(\partial_t T + b \cdot \nabla T) - \operatorname{div}(k\nabla T) = Q$$

Advection: initial and boundary conditions

In the following, we will focus on the non-conservative form:

$$\partial_t u(t,x) + b(x) \cdot \nabla u(t,x) = f(t,x)$$

This PDE is complemented with an initial condition

$$u(0,x) = u_0(x)$$

When the advection equation is posed on a domain $\Omega \subset \mathbb{R}^d$, boundary conditions should be added on the part of the boundary where the velocity field b enters Ω:

$$u(t,x) = g(x) \text{ for } (t,x) \in (0,T) \times \Gamma_-$$

where

$$\Gamma_- = \{x \in \partial\Omega \text{ s.t. } \quad b(x) \cdot n(x) < 0\}$$

(n is the outward unit normal to $\partial\Omega$)

The wave equation in physics

The scalar wave equation is

$$\partial_{t,t} u - c^2 \Delta u = f$$

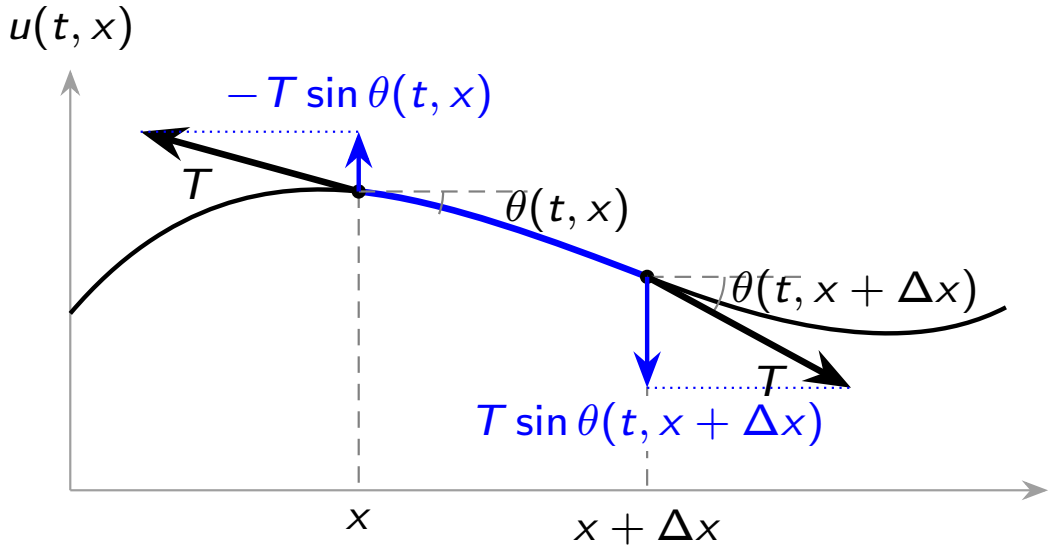
It appears in several settings

- transverse vibrations of strings and membranes
- linear acoustics: pressure and velocity perturbations in a liquid or gas at rest
- longitudinal strain waves in elastic bars
- elastic P-waves and S-waves in solids
- electromagnetic waves

In mechanics, the equation follows from Newton's law combined with a constitutive law. In electromagnetism, it follows from Maxwell's equations

A vibrating string: Newton's law

Let $u(t,x)$ be the transverse displacement of a string with mass per unit length ρ and constant tension T



Newton's law in the transverse direction for the element $[x, x+\Delta x]$ gives

$$\rho\Delta x\,\partial_{t,t}u(t,x) \simeq T\sin\theta(t,x+\Delta x) - T\sin\theta(t,x) + f(t,x)\Delta x$$

($\theta < 0$ on the figure). For small slopes, $\sin\theta \simeq \tan\theta = \partial_x u$. Dividing by Δx and sending Δx to zero gives

$$\rho\partial_{t,t}u - T\partial_{x,x}u = f$$

Longitudinal strain waves

Let $u(t, x)$ be the longitudinal displacement in an elastic bar. In the small-deformation regime, the strain is

$$\varepsilon = \partial_x u$$

Hooke's law relates the stress σ to the strain

$$\sigma = E\varepsilon = E\partial_x u$$

Newton's law gives the balance of momentum

$$\rho\partial_{t,t} u - \partial_x \sigma = f$$

Therefore

$$\partial_{t,t} u - c^2 \partial_{x,x} u = \frac{f}{\rho} \text{ with } c = \sqrt{\frac{E}{\rho}}$$

This can be generalized to homogeneous isotropic elastic materials in 3D: the dilatation and rotation satisfy wave equations (with different velocities)

Electromagnetic waves

Let $E(t,x) \in \mathbb{R}^3$ and $B(t,x) \in \mathbb{R}^3$ be the electric and magnetic fields. In a homogeneous medium without charges or currents, Maxwell's equations are

$$\operatorname{div} E = 0, \quad \operatorname{div} B = 0, \quad \partial_t B + \operatorname{curl} E = 0, \quad \operatorname{curl} B = \mu\epsilon\, \partial_t E$$

Using

$$\operatorname{curl}(\operatorname{curl} V) = \nabla(\operatorname{div} V) - \Delta V$$

one obtains

$$\partial_{t,t} E - c^2 \Delta E = 0 \text{ and } \partial_{t,t} B - c^2 \Delta B = 0 \text{ with } c = \frac{1}{\sqrt{\mu\epsilon}}$$

In vacuum, $c \simeq 3.0 \times 10^8 \ \mathrm{m.s}^{-1}$ (speed of light)

Initial and boundary conditions for waves

In the following, we focus on the 1D wave equation ($c > 0$)

$$\partial_{t,t} u(t,x) - c^2 \partial_{x,x} u(t,x) = f(t,x)$$

Since the wave equation is second order in time, two initial conditions are required:

$$u(0,x) = u_0(x), \qquad \partial_t u(0,x) = v_0(x)$$

When the equation is posed on a domain $(0, L)$, boundary conditions are needed:

- fixed endpoints: $u(t,0) = u(t,L) = 0$
- free endpoints: $\partial_x u(t,0) = \partial_x u(t,L) = 0$
- absorbing boundary conditions:

$$(\partial_t u - c\partial_x u)(t,0) = (\partial_t u + c\partial_x u)(t,L) = 0$$

Advection and characteristics

Find $u : [0, T] \times \mathbb{R}^d \to \mathbb{R}$ solution to

$$\begin{cases} \partial_t u(t,x) + b(x) \cdot \nabla u(t,x) = f(t,x) \text{ in } [0,T] \times \mathbb{R}^d \\ u(0,\cdot) = u_0 \text{ in } \mathbb{R}^d \end{cases} \tag{1}$$

where $b \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d)$, $u_0 \in \mathcal{C}^1(\mathbb{R}^d)$ and $f \in \mathcal{C}^1([0,T] \times \mathbb{R}^d)$. We consider classical solutions $u \in \mathcal{C}^1([0,T] \times \mathbb{R}^d)$

We will derive a representation formula for the solution using the flow of the ODE associated with b, and then discuss qualitative properties of this solution

Backward characteristics

Let $b \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d)$ and denote by $\phi(x_0, t)$ the flow of the ODE

$$\dot{X}(t) = b(X(t)), \qquad X(0) = x_0$$

We assume that the flow ϕ is globally defined on $\mathbb{R}^d \times \mathbb{R}$

The backward characteristic ending at x at time t is

$$s \in [0, t] \mapsto \phi(x, s - t)$$

One can check that

$$\frac{d}{ds} u(s, \phi(x, s - t)) = f(s, \phi(x, s - t))$$

Integrating along the backward characteristic, from $s = 0$ to $s = t$, gives

$$u(t, x) = u_0(\phi(x, -t)) + \int_0^t f(s, \phi(x, s - t))\, ds$$

Representation formula for advection

Theorem

Assume that $b \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d)$ *has a global flow,* $u_0 \in \mathcal{C}^1(\mathbb{R}^d)$ *and* $f \in \mathcal{C}^1([0,T] \times \mathbb{R}^d)$*. The unique classical solution to* (1) *is:*

$$u(t,x) = u_0(\phi(x,-t)) + \int_0^t f(s, \phi(x, s-t))\, ds$$

Remark: When $b(x) = a \in \mathbb{R}^d$ is constant, the flow is $\phi(x_0, t) = x_0 + at$ and the solution is thus:

$$u(t,x) = u_0(x - at) + \int_0^t f\big(s, x - a(t-s)\big)\, ds$$

Qualitative properties of advection

For $f = 0$,

$$u(t, x) = u_0(\phi(x, -t))$$

Consequently

- positivity and lower and upper bounds are preserved
- $\operatorname{supp} u(t, \cdot) = \phi(\operatorname{supp} u_0, t)$
- regularity is transported but no new regularity is created

By the change of variable $x = \phi(x_0, t)$,

$$\|u(t, \cdot)\|_{L^2}^2 = \int_{\mathbb{R}^d} |u_0(x_0)|^2 j(x_0, t)\, dx_0$$

Moreover, from Liouville's formula, if $|\operatorname{div} b| \leq L$, then $e^{-Lt} \leq j(\cdot, t) \leq e^{Lt}$ and thus

$$e^{-Lt/2}\|u_0\|_{L^2} \leq \|u(t, \cdot)\|_{L^2} \leq e^{Lt/2}\|u_0\|_{L^2}$$

In particular, the L^2 norm is conserved when $\operatorname{div} b = 0$

Liouville's formula: evolution of the jacobian of the flow

Let

$$J(x_0,t) = \nabla_{x_0}\phi(x_0,t) \in \mathbb{R}^{d\times d} \qquad \text{and} \qquad j(x_0,t) = \det J(x_0,t)$$

Differentiating $\partial_t\phi(x_0,t) = b(\phi(x_0,t))$ with respect to x_0 gives

$$\partial_t J(x_0,t) = \nabla b(\phi(x_0,t))J(x_0,t), \qquad J(x_0,0) = \mathrm{Id}$$

Jacobi's formula (which is a consequence of $\det(\mathrm{Id}+\epsilon M) = 1+\epsilon\mathrm{tr}(M)+O(\epsilon^2)$) yields

$$\begin{aligned}\partial_t j(x_0,t) &= j(x_0,t)\,\mathrm{tr}\big(J(x_0,t)^{-1}\partial_t J(x_0,t)\big)\\ &= j(x_0,t)\,\mathrm{tr}\big(J(x_0,t)^{-1}\nabla b(\phi(x_0,t))J(x_0,t)\big)\\ &= j(x_0,t)\,\mathrm{div}\; b(\phi(x_0,t))\end{aligned}$$

Therefore

$$j(x_0,t) = \exp\left(\int_0^t \mathrm{div}\; b(\phi(x_0,s))\,ds\right)$$

Wave equation

Find $u : [0, T] \times \mathbb{R} \to \mathbb{R}$ solution to

$$\begin{cases} \partial_{t,t} u(t,x) - c^2 \partial_{x,x} u(t,x) = f(t,x) \\ u(0,x) = u_0(x), \qquad \partial_t u(0,x) = v_0(x) \end{cases} \tag{2}$$

where $c > 0$, $u_0 \in \mathcal{C}^2(\mathbb{R})$, $v_0 \in \mathcal{C}^1(\mathbb{R})$ and $f \in \mathcal{C}^{0,1}([0, T] \times \mathbb{R})$.
We consider classical solutions $u \in \mathcal{C}^2([0, T] \times \mathbb{R})$

We first derive an integral representation of the solution (d'Alembert's formula). We then discuss qualitative properties of the solution

Traveling waves and characteristic variables

Let us first consider $f = 0$:

$$\partial_{t,t} u - c^2 \partial_{x,x} u = 0$$

Introduce the change of variable

$$\xi = x+ct, \qquad \eta = x-ct, \qquad U(\xi, \eta) \text{ s.t. } u(t,x) = U(x+ct, x-ct)$$

The chain rule gives

$$\partial_{t,t} u(t,x) - c^2 \partial_{x,x} u(t,x) = -4c^2 \partial_{\xi,\eta} U(x+ct, x-ct)$$

Hence $\partial_{\xi,\eta} U = 0$, i.e. $U(\xi, \eta) = F(\xi) + G(\eta)$ and thus

$$u(t,x) = F(x+ct) + G(x-ct)$$

Every solution is the superposition of a left-going and a right-going traveling wave

D'Alembert's formula

Theorem

For $f = 0$, if $u_0 \in \mathcal{C}^2(\mathbb{R})$ and $v_0 \in \mathcal{C}^1(\mathbb{R})$, the unique classical solution to (2) is

$$u(t,x) = \frac{u_0(x-ct) + u_0(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} v_0(y)\,dy$$

Proof: $u(t,x) = F(x+ct) + G(x-ct)$. Thus,

$$F(x) + G(x) = u_0(x)$$

$v_0(x) = cF'(x) - cG'(x)$ i.e.

$$F(x) - G(x) = \frac{1}{c}\int_0^x v_0(y)\,dy + C$$

Therefore, $F(\xi) = \frac{1}{2}u_0(\xi) + \frac{1}{2c}\int_0^\xi v_0(y)\,dy + \frac{C}{2}$ and $G(\eta) = \frac{1}{2}u_0(\eta) - \frac{1}{2c}\int_0^\eta v_0(y)\,dy - \frac{C}{2}$

D'Alembert's formula with a source term

Let us now consider $f \neq 0$:

$$\partial_{t,t} u - c^2 \partial_{x,x} u = f$$

The solution to (2) is (check it!)

$$u(t,x) = \frac{u_0(x-ct) + u_0(x+ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} v_0(y)\, dy + \frac{1}{2c} \int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(s,y)\, dy\, ds$$

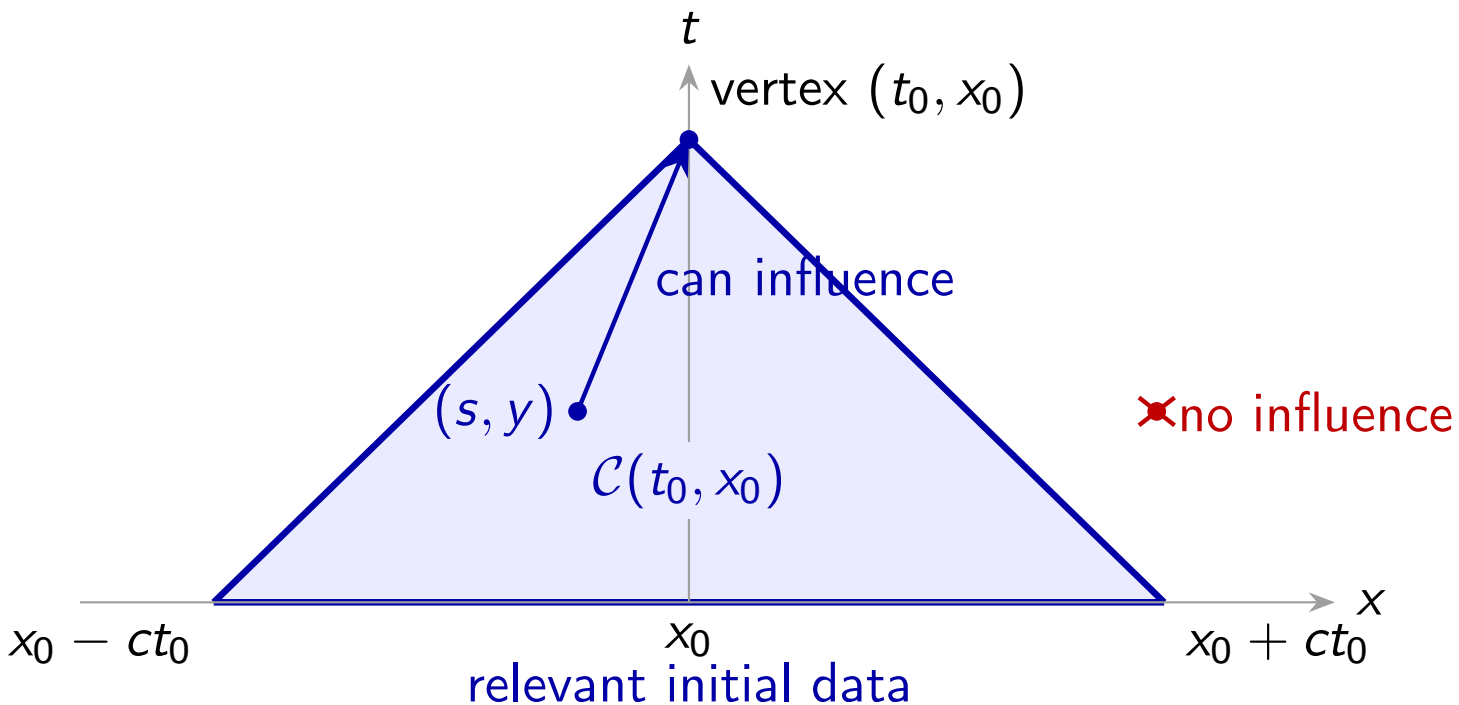
This is Duhamel's principle: at every time s, the forcing acts as a new initial-velocity contribution

Finite speed of propagation (1/2)

For any $(t_0, x_0) \in \mathbb{R}_+^* \times \mathbb{R}$, the backward cone is defined by:

$$\mathcal{C}(t_0, x_0) = \{(t, x) : 0 \leq t \leq t_0, |x - x_0| \leq c(t_0 - t)\}$$

The value of $u(t_0, x_0)$ only depends on the data (u_0, v_0, f) within the backward cone $\mathcal{C}(t_0, x_0)$



Finite speed of propagation (2/2)

Proposition

Let u and $\widetilde{u}$ be classical solutions of

$$\partial_{t,t} u - c^2 \partial_{x,x} u = f, \qquad \partial_{t,t} \widetilde{u} - c^2 \partial_{x,x} \widetilde{u} = \widetilde{f}$$

If

- $u(0,\cdot) = \widetilde{u}(0,\cdot)$ *on* $[x_0 - ct_0, x_0 + ct_0]$
- $\partial_t u(0,\cdot) = \partial_t \widetilde{u}(0,\cdot)$ *on* $[x_0 - ct_0, x_0 + ct_0]$
- $f = \widetilde{f}$ *in* $\mathcal{C}(t_0, x_0)$

then

$$u = \widetilde{u} \quad \text{in } \mathcal{C}(t_0, x_0)$$

In particular, if $u_0 = v_0 = 0$ on $[x_0 - ct_0, x_0 + ct_0]$ and $f = 0$ in $\mathcal{C}(t_0, x_0)$ then $u = 0$ in $\mathcal{C}(t_0, x_0)$

The wave equation does not regularize

Let us assume in this slide $f = 0$.

Proposition

If $u_0 \in \mathcal{C}^k(\mathbb{R})$ and $v_0 \in \mathcal{C}^{k-1}(\mathbb{R})$, with $k \geq 2$, then the homogeneous solution belongs to $\mathcal{C}^k(\mathbb{R}_+ \times \mathbb{R})$

In general, the solution is not smoother: D'Alembert's formula only translates the initial profiles

For example, let $\psi \in \mathcal{C}^k(\mathbb{R}) \setminus \mathcal{C}^{k+1}(\mathbb{R})$ and choose

$$u_0 = \psi, \qquad v_0 = -c\psi'$$

Then

$$u(t,x) = \psi(x - ct)$$

The lack of regularity is simply transported to the right at speed c

Finite difference approximations

We now discretize the advection and wave equations in dimension one. As in the previous parts, the convergence analysis follows

$$\text{consistency} + \text{stability} \Longrightarrow \text{convergence}$$

Outline:

- Advection equation: the upwind scheme, convergence analysis, and 3 remarks
- Wave equation: centered scheme, convergence analysis

The upwind scheme

Let $a > 0$ and consider

$$\begin{cases} \partial_t u + a\partial_x u = f & \text{on } (0,T)\times(0,L) \\ u(t,0) = g(t) & \text{for } t \in (0,T) \\ u(0,x) = u_0(x) & \text{for } x \in (0,L) \end{cases}$$

Set $t_n = n\Delta t$, $N = T/\Delta t$, $x_j = jh$, $J = L/h$ and $u(t_n, x_j) \simeq U_j^n$

Since information travels from left to right, use the backward difference: for $j \in \{1,\dots,J\}$ and $n \in \{0,\dots,N-1\}$

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a\frac{U_j^n - U_{j-1}^n}{h} = f(t_n, x_j)$$

with $U_j^0 = u_0(x_j)$ and $U_0^n = g(t_n)$. Using the Courant number $\nu = a\Delta t/h$, the recurrence can be rewritten as

$$U_j^{n+1} = (1-\nu)U_j^n + \nu U_{j-1}^n + \Delta t\, f(t_n, x_j)$$

For $a < 0$, we would use the forward difference instead

Consistency of the upwind scheme

Insert the exact solution into the scheme and define ε_j^n by

$$\frac{u(t_{n+1}, x_j) - u(t_n, x_j)}{\Delta t} + a\frac{u(t_n, x_j) - u(t_n, x_{j-1})}{h} = f(t_n, x_j) + \varepsilon_j^n$$

Taylor expansions give

$$\frac{u(t_{n+1}, x_j) - u(t_n, x_j)}{\Delta t} = \partial_t u(t_n, x_j) + O(\Delta t)$$

$$\frac{u(t_n, x_j) - u(t_n, x_{j-1})}{h} = \partial_x u(t_n, x_j) + O(h)$$

Therefore, if $u \in \mathcal{C}^2([0, T] \times [0, L])$,

$$\max_{n,j} |\varepsilon_j^n| \leq C(\Delta t + h)$$

Discrete maximum principle and stability

Assume the Courant–Friedrichs–Lewy (CFL) condition

$$0 \leq \nu = \frac{a\Delta t}{h} \leq 1$$

Under the CFL condition, for $f = 0$,

$$U_j^{n+1} = (1-\nu)U_j^n + \nu U_{j-1}^n$$

is a convex combination

Proposition

Assume $f = 0$. Then, for every $n \in \{0, \ldots, N\}$ and $j \in \{0, \ldots, J\}$,

$$\min\left\{\min_{[0,L]} u_0, \min_{[0,t_n]} g\right\} \leq U_j^n \leq \max\left\{\max_{[0,L]} u_0, \max_{[0,t_n]} g\right\}$$

In particular, if $u_0 \geq 0$ and $g \geq 0$, then $U_j^n \geq 0$

In particular, if $f = g = 0$,

$$\|U^{n+1}\|_\infty \leq \|U^n\|_\infty$$

Thus the upwind scheme is stable in l^∞ under the CFL condition

Convergence of the upwind scheme

Remember

$$u(t_{n+1}, x_j) = (1-\nu)u(t_n, x_j) + \nu u(t_n, x_{j-1}) + \Delta t\, f(t_n, x_j) + \Delta t\, \varepsilon_j^n$$

$$U_j^{n+1} = (1-\nu)U_j^n + \nu U_{j-1}^n + \Delta t\, f(t_n, x_j)$$

We thus obtain the recurrence for the error $E_j^n = u(t_n, x_j) - U_j^n$:

$$E_j^{n+1} = (1-\nu)E_j^n + \nu E_{j-1}^n + \Delta t \varepsilon_j^n$$

Under the CFL condition $\nu \in [0,1]$, and thus (using $E_0^n = 0$)

$$\|E^{n+1}\|_\infty \leq \|E^n\|_\infty + \Delta t \|\varepsilon^n\|_\infty$$

Hence (using $E_j^0 = 0$)

$$\max_{0\leq n\leq N} \|E^n\|_\infty \leq T \max_{0\leq n\leq N-1} \|\varepsilon^n\|_\infty$$

Theorem

Under the CFL condition $a\Delta t < h$, for a $\mathcal{C}^2$ solution, the upwind scheme is convergent of order 1: $\max_{0\leq n\leq N} \|E^n\|_\infty \leq CT(\Delta t + h)$

Remark 1: The downwind scheme is unstable

For $a > 0$, using the forward difference gives the consistent discretization

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a\frac{U_{j+1}^n - U_j^n}{h} = 0$$

i.e.

$$U_j^{n+1} = (1+\nu)U_j^n - \nu U_{j+1}^n$$

This is not a convex combination. For the oscillatory initial condition $U_j^0 = (-1)^j$,

$$U_j^n = (1+2\nu)^n(-1)^j$$

The scheme is unstable for every $\nu > 0$

The numerical derivative must use the values on the side from which the information arrives: the upwind direction

Take-home message: consistency alone does not imply convergence

Remark 2: A centered explicit scheme is unstable

The centered scheme is

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a\frac{U_{j+1}^n - U_{j-1}^n}{2h} = 0,$$

i.e.

$$U_j^{n+1} = U_j^n - \frac{\nu}{2}\left(U_{j+1}^n - U_{j-1}^n\right)$$

It is consistent, but consider a discrete oscillation

$$U_j^n = G^n \exp(\mathrm{i}j\theta)$$

The amplification factor is

$$G = 1 - \mathrm{i}\nu \sin\theta, \qquad |G|^2 = 1 + \nu^2 \sin^2\theta$$

For every $\nu \neq 0$, some modes satisfy $|G| > 1$ and grow at every time step

Remark 3: Upwind equals centered plus diffusion

The backward difference satisfies the exact identity

$$\frac{U_j^n - U_{j-1}^n}{h} = \frac{U_{j+1}^n - U_{j-1}^n}{2h} - \frac{h}{2}\frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{h^2}$$

Therefore the upwind scheme can be written

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a\frac{U_{j+1}^n - U_{j-1}^n}{2h} = \frac{ah}{2}\frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{h^2}$$

The upwind scheme is exactly a centered scheme with a positive numerical diffusion term

A centered scheme for the wave equation

Consider

$$\begin{cases} \partial_{t,t} u - c^2 \partial_{x,x} u = f \quad \text{on } (0,T)\times(0,L) \\ u(t,0) = u(t,L) = 0 \\ u(0,x) = u_0(x), \quad \partial_t u(0,x) = v_0(x) \end{cases}$$

Centered differences in time and space give

$$\frac{U_j^{n+1} - 2U_j^n + U_j^{n-1}}{\Delta t^2} - c^2 \frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{h^2} = f(t_n, x_j)$$

where $n \in \{1, \dots, N-1\}$ with $N = T/\Delta t$ and $j \in \{1, \dots, I-1\}$ with $I = L/h$. The scheme can be written

$$U^{n+1} - 2U^n + U^{n-1} = -c^2 \Delta t^2 A U^n + \Delta t^2 F^n$$

where A is the matrix of the finite difference approximation of $-\partial_{x,x}$, and $F_j^n = f(t_n, x_j)$. (We consider for simplicity homogeneous boundary conditions)

Initialization and matrix form

Reminder:

$$A = \frac{1}{h^2}\begin{bmatrix} 2 & -1 & & 0 \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ 0 & & -1 & 2 \end{bmatrix} \in \mathbb{R}^{(I-1)\times(I-1)}$$

The recurrence is explicit but uses two time levels. We therefore need an initialization for U^0 and U^1:

$$U_j^0 = u_0(x_j)$$

and, using Taylor expansion at $t = 0$, set

$$U_j^1 = u_0(x_j) + \Delta t\, v_0(x_j) + \frac{c^2\Delta t^2}{2h^2}\big(u_0(x_{j+1}) - 2u_0(x_j) + u_0(x_{j-1})\big) + \frac{\Delta t^2}{2} f(0, x_j)$$

Consistency of the wave scheme

The consistency error is defined by: for $n \in \{1, \dots, N-1\}$ and $j \in \{1, \dots, I-1\}$,

$$\frac{u(t_{n+1}, x_j) - 2u(t_n, x_j) + u(t_{n-1}, x_j)}{\Delta t^2} - c^2 \frac{u(t_n, x_{j+1}) - 2u(t_n, x_j) + u(t_n, x_{j-1})}{h^2} = f(t_n, x_j) + \varepsilon_j^n$$

The initialization error is: for $j \in \{1, \dots, I-1\}$,

$$\eta_j^1 = u(t_1, x_j) - U_j^1$$

For a sufficiently regular solution, Taylor expansions give

$$\max_{n \in \{1, \dots, N-1\}} \|\varepsilon^n\|_\infty \leq C(\Delta t^2 + h^2)$$

$$\|\eta^1\|_\infty \leq C(\Delta t^3 + \Delta t^2 h^2)$$

Modal stability and the CFL condition

The matrix A is symmetric positive definite, with eigenvalues

$$\mu_k = \frac{4}{h^2} \sin^2\left(\frac{k\pi}{2I}\right), \qquad 1 \le k \le I-1$$

On an eigenmode, the scheme for $f = 0$ becomes

$$y^{n+1} - (2 - c^2\Delta t^2 \mu) y^n + y^{n-1} = 0$$

The roots of the characteristic polynomial have modulus one when

$$0 < c^2\Delta t^2 \mu < 4$$

Since $\mu_k < 4/h^2$, every mode is stable under the CFL condition

$$\frac{c\Delta t}{h} \le 1$$

If it is violated (and h is sufficiently small), the highest-frequency modes grow exponentially

Stability in l^2

Define the discrete initial velocity

$$V^0 = \frac{U^1 - U^0}{\Delta t} + \frac{c^2 \Delta t}{2} A U^0$$

Lemma

Under the CFL condition $c\Delta t/h \leq 1$,

$$\max_{m \in \{0,\ldots,n\}} \|U^m\|_2 \leq \|U^0\|_2 + \frac{L}{2c} \left(\|V^0\|_2 + \Delta t \sum_{m=1}^{n-1} \|F^m\|_2 \right)$$

The proof decomposes U^n on an orthonormal eigenbasis of A and solves the scalar recurrence for each mode

The scheme is stable in l^2 under the CFL condition

Convergence of the wave scheme

Define the global error

$$E_j^n = u(t_n, x_j) - U_j^n$$

It satisfies the same recurrence as the numerical solution, with source ε^n

$$E^{n+1} - 2E^n + E^{n-1} = -c^2 \Delta t^2 A E^n + \Delta t^2 \varepsilon^n,$$

and $E^0 = 0$, $E^1 = \eta^1$

Theorem

For a sufficiently regular solution, if $c\Delta t/h \leq 1$, then

$$\max_{n \in \{0,\dots,N\}} \|E^n\|_2 \leq C(\Delta t^2 + h^2)$$

The scheme has order two in time and space

Remark: It can be shown that if $c\Delta t/h \leq 1 - \delta$, then $\max_{n\in\{0,\dots,N\}} \|E^n\|_\infty \leq C_\delta(\Delta t^2 + h^2)$

Comparison of the CFL conditions

For the explicit schemes studied in these lectures, the stability conditions have different scalings

- heat equation: $\Delta t \leq Ch^2$
- advection equation: $|a|\Delta t \leq h$
- wave equation: $c\Delta t \leq h$

For transport and waves, the condition has a geometric interpretation: The numerical domain of dependence must contain the domain of dependence of the differential equation. During one time step, information should not travel across more than approximately one mesh cell.

For diffusion, the more restrictive h^2 scaling reflects the large eigenvalues of the discrete Laplacian and the fact that the explicit Euler method must damp every spatial mode